

CHRISTOPHER FU

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EDUCATION

University of California, Los Angeles (UCLA)

M.S. Data Science in Biomedicine

Expected Graduation Date: June 2026

Overall GPA: 4.00/4.00

Relevant Coursework: Foundations of Data Science, Machine Learning Applications in Biomedicine, Data Science for Medical Imaging

Awards: University of California, Los Angeles Warren Alpert Computational Biology and Artificial Intelligence Scholar, Dean's Honor List

University of California, Los Angeles (UCLA)

B.S. Data Theory

Graduation Date: June 2024

Overall GPA: 3.79/4.00

PROFESSIONAL EXPERIENCE

Warren Alpert Computational Biology and AI Network

Los Angeles, CA

Warren Alpert Computational Biology and AI Network Fellow

Sep. 2025 - Dec. 2025

- Built an end-to-end pipeline combining longitudinal post-mortem MRI data, SynthSeg, and autopsy reports to quantify brain structure volumes and deformation rates across patients, linking imaging-based biomarkers with ground truth labels for neurodegeneration analysis
- Developed scalable analysis workflows including 3D segmentation validation, patient-level heatmaps of anatomical volumes and deformation trends, and a pair plot of autopsy variables to assess feature relationships, producing a dataset of 1300 rows for our cohort
- Applied LOOCV logistic regression to evaluate performance of brain structures in Alzheimer's Disease classification, achieving 65% accuracy with high F1 on select features while underscoring sparse, interpretable biomarkers and limitations of small, imbalanced cohorts

Kaiser Permanente

Pasadena, CA

Applied Data and Analytics Master's Intern

Jun. 2025 - Sep. 2025

- Performed EDA and feature engineering in Python on 1K physician survey responses, identifying demographic and event-level drivers of engagement; partnered with the CME research consultants to improve survey design and integrate findings into a future AI Summit
- Fine-tuned Hugging Face transformer models (Few-Shot learning) on 950K dermatology patient messages, collaborating with a dermatologist to build a message classifier for triage (Macro F1 = 0.891) ready for Fall 2025 deployment to improve clinician workflows
- Adapted Kaiser Permanente's primary care message classification model across 5 medical specialties, identifying low-volume underperforming classes; provided recommendations to enhance specialty-level classification accuracy for predicting patient messages

UCLA Biodesign

Los Angeles, CA

Biodesign AI Fellow

Sep. 2024 - Jun. 2025

- Leveraged SQL to process, clean, and integrate OB/GYN patient flow data from a dataset of 80,956 rows spanning 2022–2024; conducted exploratory analysis using visualizations to uncover trends in bed usage, delivery patterns, and departmental efficiency metrics
- Designed, implemented, and evaluated an LSTM model to forecast OB capacity time-series trends, training on 2022–2023 data; achieved a test RMSE of 17.69, providing actionable insights to optimize resource and staff allocation, as well as enhance operational planning
- Partnered with a team of six to analyze patient flow data within a school's comprehensive data repository; developed a project strategy, presented findings, and proposed actionable next steps, emphasizing opportunities to improve efficiency and data-driven decisions

Children's Hospital of Orange County (CHOC)

Orange, CA

Data Science Intern

Jun. 2023 - Aug. 2023

- Analyzed demographic disparities among diagnosed anxiety patients by constructing multiple linear regression models using Python's Statsmodels and SciPy libraries, achieving 77% accuracy; conducted t-tests and other statistical analyses to identify actionable insights
- Led a team of four data science interns in delivering data-driven insights to physicians through visually engaging presentations; created bar plots, box plots, model summary tables, and geographical heat maps to highlight trends and promote predictive measures in mental health
- Developed and proposed detailed action plans for deploying advanced unsupervised machine learning techniques to uncover latent factors contributing to anxiety disorders in pediatric patients, supporting data-driven interventions and improved outcomes

SKILLS

Programming: Python (NumPy, Matplotlib, SciKit-Learn, Pandas, PyTorch, Jupyter, SciPy), R Studio (ggplot2), SQL (Azure Data Studio)

Data Analysis & Machine Learning: Regression Analysis, Classification Models, Unsupervised Learning, Neural Networks (LSTM, VAE)

BI & Visualization Tools: Tableau, PowerBI, MS Excel, Google Sheets, MS Powerpoint, MS Word